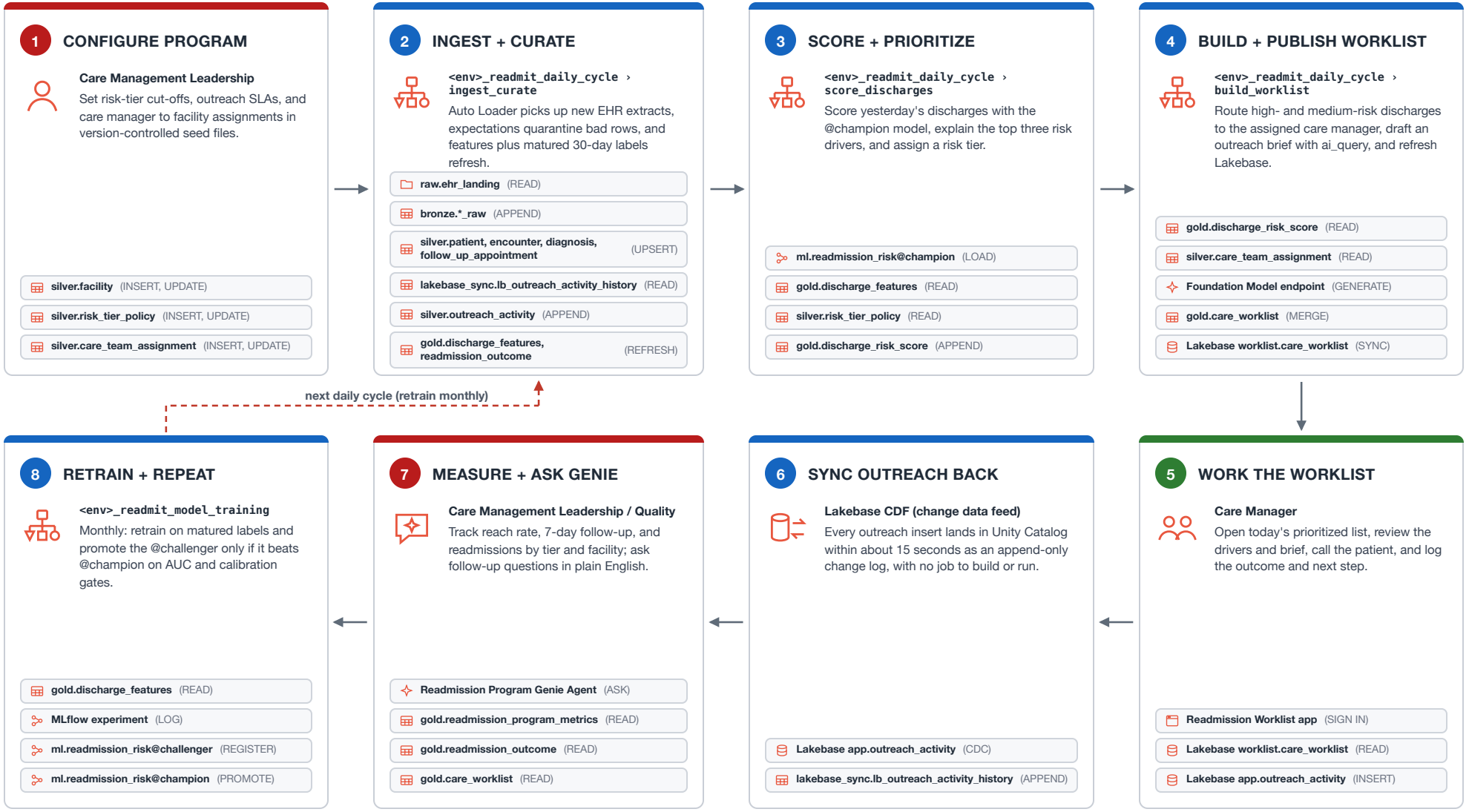
Readmission Prevention Worklist — Recommended Operating Pattern

Draft v0.2 • November 2, 2026

Eight operational steps from the nightly EHR extract to care manager outreach and measured 30-day outcomes, with the tables and assets touched in each step.

Deployment: workspace <env>_clinical_analytics • catalog <env>_care_management • Lakebase project <env>-care-management



READMISSION WORKLIST APP

Care managers work their list and log outreach; leadership uses the Program tab with the embedded Genie Agent. The app reads and writes Lakebase, and Genie runs on behalf of the signed-in user.

GOVERNANCE (EVERY STEP)

PHI columns carry governed tags. Tag-driven ABAC policies mask direct identifiers and filter rows by facility for every SQL, notebook, and Genie read. The Lakebase CDF landing schema stays unmasked (a CDF requirement), so only the pipeline identity can read it; policies apply from silver onward.

Ownership: red = Care Management Leadership • green = Care Manager • blue = automated Data Platform job or managed sync

Actions: READ = read only • INSERT = add rows • UPDATE = change rows • APPEND = append-only insert • UPSERT = AUTO CDC apply • MERGE = keyed upsert • REFRESH = materialized view recompute • LOAD / REGISTER / PROMOTE = UC model alias operations • LOG = MLflow run • GENERATE = ai_query call • SYNC = Delta to Lakebase synced table • CDC = Lakebase CDF to Unity Catalog • ASK = Genie question • SIGN IN = authenticated app session

Readmission Prevention Worklist — Environment-Local Deployment Pattern

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One Declarative Automation Bundle deploys every asset per environment; each workspace reads and writes only its own <env> catalog and Lakebase project.

Catalog Design Pattern

UNITY CATALOG CATALOG
<env>_care_management

READ/WRITE binding

- SCHEMA raw — Volume: ehr_landing
- SCHEMA bronze — patient_raw, encounter_raw, diagnosis_raw, appointment_raw, quarantine_raw
- SCHEMA silver — patient, encounter, diagnosis, follow_up_appointment, outreach_activity, facility, care_team_assignment, risk_tier_policy
- SCHEMA gold — discharge_features, discharge_risk_score, care_worklist, readmission_outcome, readmission_program_metrics (metric view)
- SCHEMA ml — Model: readmission_risk (@champion, @challenger)
- SCHEMA lakebase_sync — lb_outreach_activity_history (Lakebase CDF; restricted, no masks)
- SCHEMA governance — fn mask_direct_identifier, fn mask_birth_date, fn filter_by_facility, ABAC policies on tag phi

PHI tagging rule

Every PHI column carries governed tag **phi = direct_identifier | quasi_identifier**; ABAC policies attached to the silver and gold schemas apply masks and the facility row filter automatically to new tables. lakebase_sync is excluded because CDF destinations must not carry masks or filters.

Minimum-necessary rule

Only worklist columns a care manager needs to make the call leave the lakehouse; Lakebase never holds diagnosis history or the full chart.

Workspace Design Pattern

DATABRICKS WORKSPACE
<env>_clinical_analytics

- LAKEFLOW Pipeline — <env>_readmit_pipeline (serverless SDP)
- LAKEFLOW Jobs — <env>_readmit_daily_cycle: ingest_curate → score_discharges → build_worklist; <env>_readmit_model_training (monthly); <env>_readmit_setup (synced table + CDF + Genie, idempotent); <env>_readmit_source_simulator (demo only)
- LAKEBASE PROJECT <env>-care-management — branch production / db care_mgmt; schema worklist (synced from gold, read-only): care_worklist; schema app (app-owned, REPLICA IDENTITY FULL): outreach_activity
- DATABRICKS APP <env>-readmission-worklist — resources: postgres, sql_warehouse, genie_space; Worklist tab; Program tab
- GENIE AGENT Readmission Program — gold.readmission_program_metrics; gold.readmission_outcome; gold.care_worklist
- SERVING Foundation Model — ${var.llm_endpoint} (pay-per-token); SQL warehouse <env>_readmit_wh

App access rule

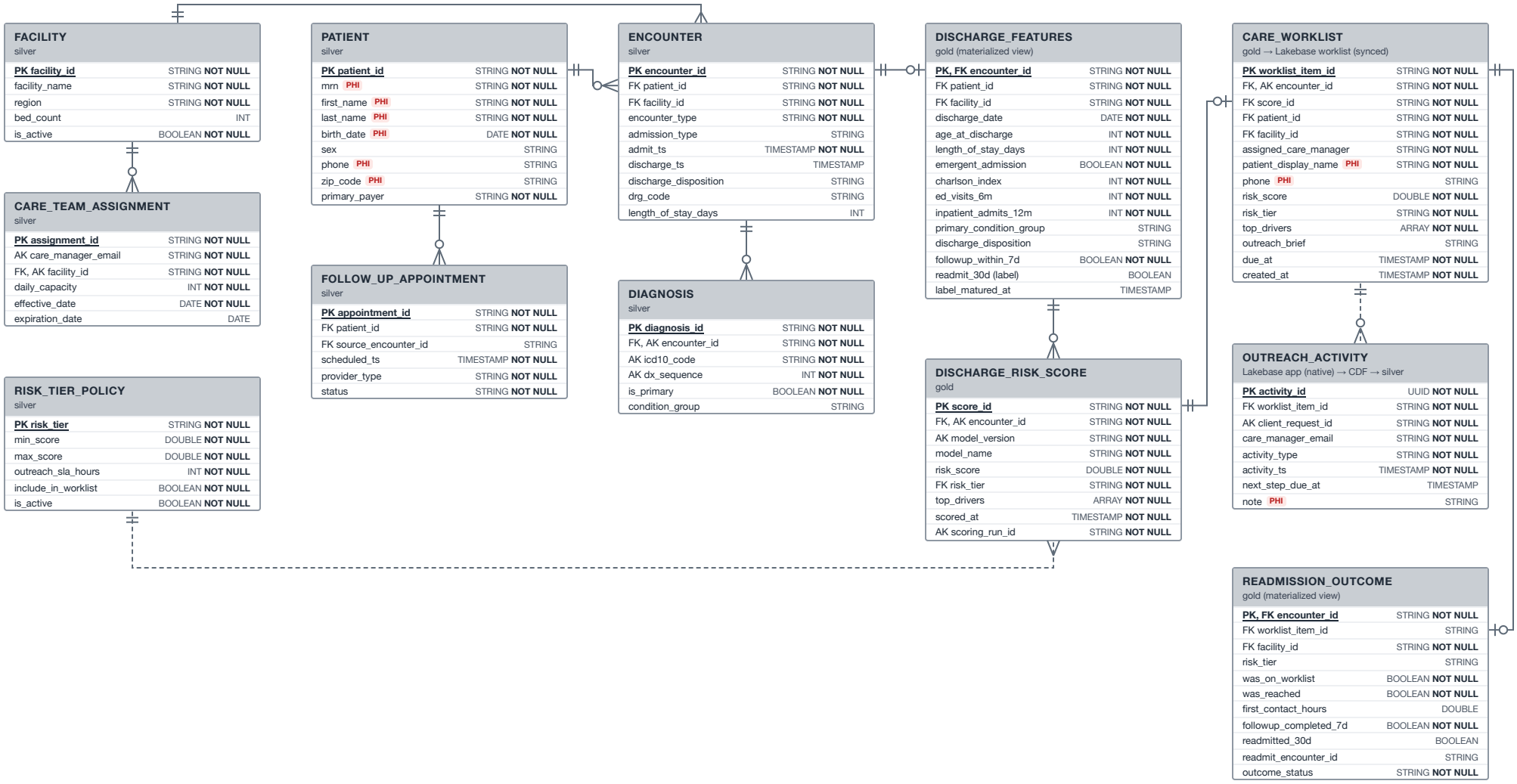
UC row filters do not reach Postgres, so every app query is scoped to the signed-in user's facility assignments; Genie runs on behalf of the user so UC policies apply.

Environment guardrail

Bundle targets dev and prod set <env>; jobs reject any catalog or Lakebase target whose prefix does not match the job's <env> parameter.

Readmission Worklist Entity-Relationship Diagram

Tables deployed in <env>_care_management (silver, gold) and Lakebase care_mgmt; recommended logical keys and workflow relationships.



Legend

- 1 to zero-or-many
- 1 to zero-or-one
- Logical reference (across Lakebase)

PK primary key • FK foreign key • AK alternate key

Status, tier, type, disposition, and payer columns are controlled STRINGs. PHI marks columns carrying governed tag phi; ABAC policies mask them for non-care-team readers.

PK/FK are Unity Catalog informational constraints (not enforced); required columns are NOT NULL; jobs enforce uniqueness, idempotency, and referential checks. AK columns are hashed to generate the PK.

Every silver and gold table also carries audit_created_at / _by / _process / _run_id and audit_updated_* columns (omitted above for readability).

gold.readmission_program_metrics (metric view over readmission_outcome) defines readmission_rate, reach_rate, followup_7d_rate, and est_cost_avoided once, for Genie and the app.